REFINED VALUATIVE CRITERION FOR SEPARATEDNESS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5189-5206, following the statement at 5170-5187.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Refined valuative criterion for properness. lemma-refined-valuative-criterion-proper (same chapter, card L26303). Applied to U -> X and the diagonal Delta: X -> X x_Y X; at-most-one dotted arrows give existence via a = b.

Source reference. Original lemma 0CMD: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 5105–5105. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ and $h : U \to X$ be morphisms of algebraic spaces over S. Assume that Y is locally Noetherian, that f is locally of finite type and quasi-separated, that h is of finite type, and that the image of $|h| : |U| \to |X|$ is dense in $|X|$. If given any commutative solid diagram*

$$\begin{array}{ccccc} \mathrm{Spec}(K) & \longrightarrow & U & \xrightarrow{h} & X \\ \downarrow & \nearrow & & & \downarrow{\scriptstyle f} \\ \mathrm{Spec}(A) & & \longrightarrow & & Y \end{array}$$

where A is a discrete valuation ring with field of fractions K, there exists at most one dotted arrow making the diagram commute, then f is separated.

Proof. We will apply Lemma 0CMD to the morphisms $U \to X$ and $\Delta : X \to X \times_Y X$. We check the conditions. Observe that Δ is quasi-compact because f is quasi-separated. Of course Δ is locally of finite type and separated (true for any diagonal morphism). Finally, suppose given a commutative solid diagram

$$\begin{array}{ccccc} \mathrm{Spec}(K) & \longrightarrow & U & \xrightarrow{h} & X \\ \downarrow & & & \nearrow & \downarrow{\scriptstyle \Delta} \\ \mathrm{Spec}(A) & & \xrightarrow[(a,b)]{} & & X \times_Y X \end{array}$$

where A is a discrete valuation ring with field of fractions K. Then a and b give two dotted arrows in the diagram of the lemma and have to be equal. Hence as dotted arrow we can use $a = b$ which gives existence. This finishes the proof. □

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.